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1. A player has sustained a severe injury during football practice and was told that he has a torn knee cartilage. Can he expect a quick uneventful recovery? Explain your response.

2. An overweight high school student, is overheard telling her friend that she is going to research how she can transform some of her white fat to brown fat. What is her rationale here (assuming it is possible)?

3.5 Neural Tissue

Nervous tissue exerts the greatest control over the body's responsiveness to changing conditions. Neurons, the unit of neural system are excitable cells (Figure 3.8). The neuroglial cells which constitute the rest of the neural system protect and support the neurons. Neuroglia makes up more than one-half of the volume of neural tissue in our body.

When a neuron is suitably stimulated, an electrical disturbance is generated which swiftly travels along its plasma membrane. Arrival of the disturbance at the neuron's endings, or output zone, triggers events that may cause stimulation or inhibition of adjacent neurons and other cells (You will study in detail in Chapter 10)

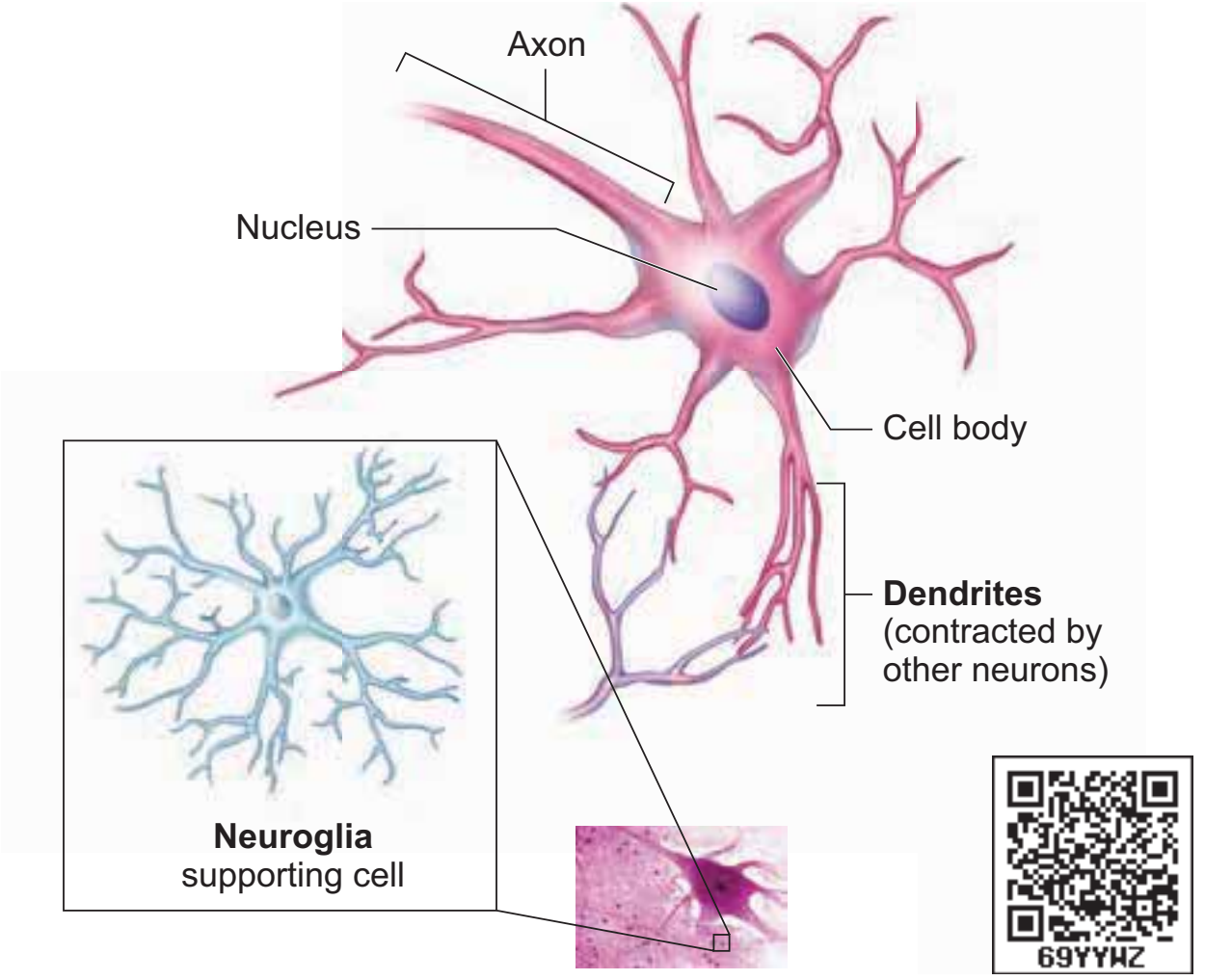


Figure 3.8 Nervous tissues with neuroglia